A Binary Characterization of OEIS A195608

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Abstract

We characterize every index at which the sum of the first n evil numbers equals one quarter of the product of the nth and $(n+1)$st evil numbers. The binary digit parity of two adjacent integers decides membership. This proves the three structural conjectures in OEIS A195608: the gaps are exactly $1, 4, 6$, the maximal consecutive blocks have length one or three, and each block start other than the initial index has the stated neighboring-evil-number difference equal to five. The initial index is an explicit boundary exception to the wording of the third conjecture, since the referenced evil-number sequence starts at index one.

1 An exact membership criterion

Let $t(k)$ be the parity of the number of ones in the binary expansion of $k \geq 0$. Then

$$t(2k) = t(k), \qquad t(2k+1) = 1 - t(k).$$

Write $E(n)$ for the nth nonnegative evil number, using the one-based indexing of OEIS A001969 [2]. Exactly one of $2k$ and $2k+1$ has even binary digit sum, so

$$E(n) = 2(n-1) + t(n-1) \qquad (n \geq 1). \tag{1}$$

Let $\mathcal{S}$ consist of the positive integers n satisfying

$$\sum_{j=1}^{n} E(j) = \frac{E(n)E(n+1)}{4}. \tag{2}$$

Its increasing enumeration is OEIS A195608 [1].

Theorem 1. *For every $k \geq 0$,*

$$2k+1 \in \mathcal{S} \iff t(k) = 0.$$

For every $k \geq 1$,

$$2k \in \mathcal{S} \iff t(k-1) = t(k) = 0.$$

Thus the entire set, with no search for the original summatory identity, is

$$\mathcal{S} = \{2k+1 : k \geq 0,\ t(k) = 0\} \cup \{2k : k \geq 1,\ t(k-1) = t(k) = 0\}.$$

Proof. Put $T(n) = \sum_{j=0}^{n-1} t(j)$. Pairing consecutive terms gives $T(2k) = k$ and $T(2k+1) = k + t(k)$. Equation (1) gives

$$\sum_{j=1}^{n} E(j) = n(n-1) + T(n).$$

For $n = 2k+1$, put $u = t(k)$. Then $t(n-1) = u$, $t(n) = 1-u$, and subtracting four times the left side of (2) from its right-hand product gives

$$E(n)E(n+1) - 4\sum_{j=1}^{n} E(j) = -2u.$$

This vanishes precisely when $u = 0$.

For $n = 2k$, put $u = t(2k-1) = 1 - t(k-1)$ and $v = t(2k) = t(k)$. The corresponding difference is

$$(4k-2)v + 4ku + uv - 4k.$$

For $(u, v) = (0,0), (0,1), (1,0), (1,1)$ its values are, respectively, $-4k, -2, 0, 4k-1$. Since $k \geq 1$, only $(u, v) = (1, 0)$ works, giving the stated condition. □

2 All three structural questions

OEIS A195608 lists three conjectures [1]: its first differences take only the values $1, 4, 6$; its maximal consecutive blocks have size one or three; and at a block's first index m, $E(m+1) - E(m-1) = 5$. The last expression is defined only for $m \geq 2$. Our result resolves these statements with that necessary boundary qualification.

Corollary 1. *The maximal consecutive blocks in $\mathcal{S}$ have length one or three. The differences of successive terms of its increasing enumeration are exactly $1, 4, 6$. The initial block is $\{1\}$. Every later block starts at an odd integer m and satisfies*

$$E(m+1) - E(m-1) = 5.$$

Proof. Every pair $t(2r), t(2r+1)$ consists of opposite bits. Therefore neither 000 nor 111 occurs in the parity sequence. Consecutive zero positions are at distance one, two, or three. A run of zero positions has length at most two.

By Theorem 1, a single zero at k produces the odd member $2k+1$. Two adjacent zeros at $k, k+1$ produce exactly the three consecutive members $2k+1, 2k+2, 2k+3$. All even members arise in this way. Thus all maximal blocks have length one or three. Zero-position gaps of two and three produce member gaps four and six; a zero-position gap of one is filled by its even member and produces two gaps of one. All three differences occur, as the initial terms $1, 7, 11, 12, 13$ demonstrate.

A noninitial block begins at $m = 2k+1$ where $k \geq 1$, $t(k) = 0$, and $t(k-1) = 1$. Now

$$E(m+1) = 4k + 2 + t(2k+1) = 4k+3, \qquad E(m-1) = 4k - 2 + t(2k-1) = 4k-2.$$

Their difference is five. At the initial block $m = 1$, $E(0)$ is outside the domain of A001969; no assertion involving it is needed. □

3 Reproducibility and scope

The accompanying `verify.py` independently generates evil numbers by testing binary digit parity and tests the original integer equality (2). It compares this computation with the theorem for every index through one million, checks the complete first 10,000 published terms, and checks all complete blocks and block-start differences in the same interval. A second independent generation uses the binary substitution $0 \mapsto 01$, $1 \mapsto 10$. These finite audits validate the implementation; the proofs above cover every positive index.

References

[1] V. Shevelev, OEIS A195608, The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A195608, accessed October 26, 2026.

[2] The OEIS Foundation, OEIS A001969, https://oeis.org/A001969, accessed October 26, 2026.